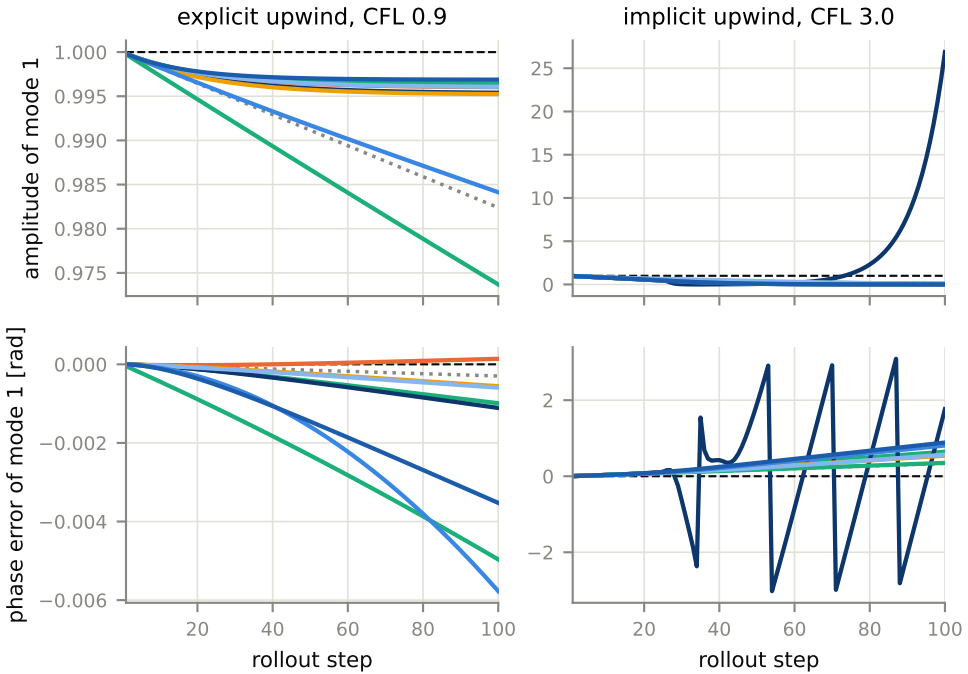FNO: rollout of a unit mode-1 wave under emulator, solver and exact dynamics